Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 1

Mandatory questions

1. (Random variables and standard deviation) A whole-life policy with sum assured £100,000 is issued to a life aged x. The sum assured is paid at the end of year of death while premiums are paid annually in advance. Let K_x denote the curtate future lifetime of x and L_0 denote the present value of the future loss on the contract, using the premium basis. Derive an expression for the standard deviation of L_0.

2. (A proof of results needed in the proof of Thiele's Differential Equation for a whole-life policy given in the notes.)
   (a) Let $V(t)$ be the net premium policy value at duration t, of a whole-life insurance sum assured £1 payable immediately on death, sold to a life age x, with level premiums payable continuously for life. Show that:

$$V(t) = 1 - \frac{\bar{a}_{x+t}}{\bar{a}_x}.$$

   (b) Show that:

$$\frac{d}{dt}\bar{a}_{x+t} = \mu_{x+t}\,\bar{a}_{x+t} - \bar{A}_{x+t}.$$

3. (Calculating net premium policy values)
   (a) Explain briefly why a life office holds reserves in respect of its policies.
   (b) A life office sells non-profit endowment assurances with term 25 years to lives age 30. It calculates net premium policy values assuming the mortality of the AM92 ultimate table. Calculate $V(5)$, $V(10)$, $V(15)$, and $V(20)$, at 4% p.a. interest.
   HINT: The numerical answers are
   $$ {}_5V_{30:\overline{25}|} = 0.1301 \quad {}_{10}V_{30:\overline{25}|} = 0.28829 \quad {}_{15}V_{30:\overline{25}|} = 0.48039 \quad {}_{20}V_{30:\overline{25}|} = 0.71399 $$
   (c) A life office sells term assurances with term 25 years to lives age 30. It calculates net premium policy values assuming the mortality of the AM92 ultimate table. Calculate $V(5)$, $V(10)$, $V(15)$, and $V(20)$, at 4% p.a. interest.
   HINT: The numerical answers are
   $$ V(5) = 0.00349 \quad V(10) = 0.00694 \quad V(15) = 0.00934 \quad V(20) = 0.00843. $$

4. (Gross premium policy values) A certain life office sold without profits term assurances to lives age 40, with term 25 years, sum assured £100,000 payable at the end of year of death and a level annual premium of £200 p.a.. It calculates policy values using the gross premium method, now on the following basis:

   mortality: AM92 ultimate

   interest: 4% per annum

   future expenses: 10% of office premiums.

   Calculate the policy value for this contract, immediately before payment of the 16th premium.

   HINT: The numerical solution is 4558.22.

5. (Supercompound bonuses) An endowment assurance with term 25 years was sold to a life aged 40, 3 years ago. The sum assured is £60,000 payable at the end of year of death and premiums are payable annually in advance. Bonuses are supercompound (two-tier) as follows: (a) each year there is a *simple* bonus of of 3% on the sum assured; and (b) an additional bonus of 4.5% on *all existing bonuses*. What is the policy value (just before payment of the 4th premium)?

   Basis:

   Method: net premium policy valuation.

   Mortality: AM92 select.

   Interest: 4% per annum.

   HINT: The numerical answer is 7062.81

6. (Excel exercise) Now go to the file "F70LB Week 1 - Excel Exercise 1 (TEMPLATE and INSTRUCTIONS)". You will find instructions for an Excel exercise there. Note that Excel skills are an important part of this course and are an opportunity to prepare yourself for the workplace. The purpose of the Excel exercise is not only to get the right answer. Make sure that you take the opportunity to practice good spreadsheet construction and presentation, and that you self-check your work. In the workplace, it is not enough to get the right answer: You will need to give your colleagues confidence that your answer is correct, make your work easy to check, and make it easy to re-use your spreadsheet.

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F70LB Tutorial 1

Optional questions

The questions in this section are all optional. They have been retained from older versions of the course to give students additional practice if they wish. Solutions will be provided, but it is not planned to cover them in tutorials. Your priority should be to cover all the main written and Excel questions before attempting any of these.

1. Note: The mortality basis A1967-70 can be downloaded via the link to the 'Green Tables' available on Canvas.

   15 years ago a life office sold a with-profits whole-life policy to a person then aged 40. The sum assured was £20,000 payable at the end of year of death and a premium of £350 was payable annually in advance. Compound reversionary bonus accrues at the start of each year. The bonus added to date is £13,000. The office values the policy on the following gross premium basis.

   Mortality: A1967–70 select.

   Interest: 6% per annum.

   Future bonuses: 2.913% p.a. (compound).

   Expenses: Nil.

   Calculate the policy value just before the 16th premium is paid.

2. A life office issues a large number of 3-year non-profit endowment assurance policies to males aged exactly 62. The sum assured under each policy is £2,000 payable at the date of maturity or at the end of year of earlier death. Under each policy, level annual premiums are payable in advance throughout the term of the policy.

   The premium and reserving bases are as follows.

   Premium basis:

   Mortality: A1967–70 select.

   Interest: 6% per annum.

   Expenses: Initial £150

   Renewal £20 at the start of the second and third policy years.

   Reserving basis: Net premium method

   Mortality: A1967–70 ultimate

   Interest: 3% per annum.

   (a) Show that the annual premium is £665.07.
   (b) Calculate the policy values per policy at durations 0, 1, and 2 years respectively.

3. (Excel exercise) You will find a file "F70LB Lab 1 - Optional Revision Exercise (TEMPLATE AND INSTRUCTIONS)" on Canvas. This repeats an exercise from F70LA, but you might find it helpful to reconsider this in light of what you have learned about good spreadsheet construction.